

Measures of Dispersion

Variance & Standard Deviation Tutorial - 5

1. What is Measure of Dispersion?

Definition: Measure of Dispersion describes how spread out or scattered the data is around the center.

Hello students! In our last lesson, we learned about Central Tendency (Mean, Median, Mode), which tells us where the **center** of the data is. Today, we are learning about **Dispersion**. If Central Tendency points to the center, Dispersion tells us how far the data *stretches* away from that center! We measure this "spread" using two super-tools: **Variance** and **Standard Deviation**.

Real-Life Example:

Imagine two cricket players, Player A and Player B. Both have an average (Mean) score of 50 runs.

- **Player A's scores:** 48, 50, 52 (Low Spread/Dispersion).

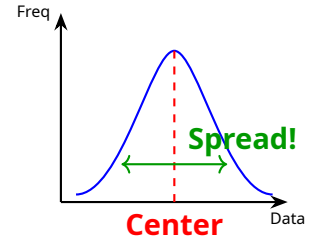
- **Player B's scores:** 0, 50, 100 (High Spread/Dispersion).

Even though their centers are the same, their *spread* is completely different!

Practice Question:

Q: If all students in a class score exactly 80 marks out of 100, what is the dispersion (spread) of the marks?

A: The dispersion is ZERO! Because the data is not spread out at all; everyone got the exact same score.



Topic Summary: Measure of Dispersion

It tells us how stretched or scattered the data is from the middle point. The two main ways to measure it are Variance and Standard Deviation.

2. Population vs. Sample Formulas

Before calculating anything, we must know if we are working with the **Population** (the whole big group) or a **Sample** (a small piece of the group). The formulas change slightly depending on which one we use!

Important Note & Extra Info:

Formula Rules:

1. Population Variance (σ^2 - 'Sigma Squared'):

We divide the squared differences by Capital **N**.

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

2. Sample Variance (s^2):

We divide the squared differences by Small **n - 1**.

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

*Why (n - 1)? This is called **Bessel's Correction** or "Degrees of Freedom". It fixes a mathematical error that happens when we use a tiny sample to guess a huge population. We will learn more about this in future videos!*

Practice Question:

Q: You want to find the variance of a Sample data set containing 10 items. What number will you divide by at the very end?

A: You will divide by $n - 1$, which is $10 - 1 = 9$.

Population

Mean = μ
Size = N

VS

Sample

Mean = $\bar{x}$
Size = n

↓
**Always divide by
n - 1 for Sample!**

3. How to Calculate Variance and Standard Deviation

Let's calculate the Sample Variance and Standard Deviation together step-by-step! Because we rarely have data for the entire world, we usually work with the *Sample* formula.

Our Data (x): 1, 2, 2, 3, 4, 5. Total count (n) = 6.

Step 1: Find the Mean ($\bar{x}$).

Sum = $1 + 2 + 2 + 3 + 4 + 5 = 17$. Mean = $17 \div 6 = 2.83$.

Step 2: Subtract Mean from each number ($x - \bar{x}$).

$1 - 2.83 = -1.83$

$2 - 2.83 = -0.83$ (do this for all numbers...)

Step 3: Square the answers to remove negatives ($(x - \bar{x})^2$).

$(-1.83)^2 = 3.34$, $(-0.83)^2 = 0.68$, and so on.

Step 4: Sum all squared values.

Total Sum ≈ 10.84 .

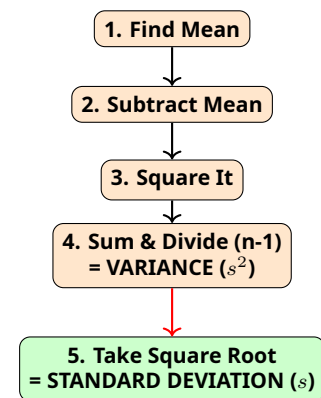
Step 5: Divide by ($n - 1$) to get VARIANCE (s^2).

Variance = $10.84 \div (6 - 1) = 10.84 \div 5 = 2.168$.

Step 6: Square Root to get STANDARD DEVIATION (s).

The final boss! Standard Deviation is just the square root of Variance.

$s = \sqrt{2.168} \approx 1.472$.



4. What does Standard Deviation mean?

So, our Standard Deviation is **1.472**. What does this number actually do? Think of Standard Deviation as the "step size" or "jump size" away from the center.

Our Mean (Center) is **2.83**.

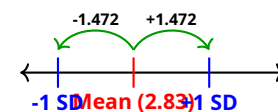
• **1 Jump Right (+1 SD):** $2.83 + 1.472 = 4.302$.

Any data between 2.83 and 4.302 lives in the "1st Standard Deviation to the right" zone!

• **2 Jumps Right (+2 SD):** $4.302 + 1.472 = 5.774$.

• **1 Jump Left (-1 SD):** $2.83 - 1.472 = 1.358$.

This is how we figure out exactly where most of our data points are hiding!



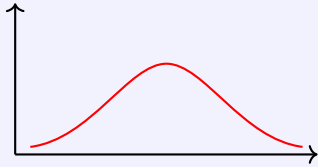
5. High Variance vs. Low Variance

How does a graph change if the variance is a huge number vs. a tiny number?

High Variance / High SD

If Variance is a **Big Number**:

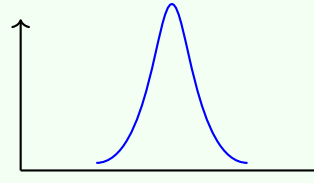
- The spread is very wide.
- The curve looks **flat and stretched**.
- The data is all over the place (inconsistent).



Low Variance / Low SD

If Variance is a **Small Number**:

- The spread is very small.
- The curve looks **tall and skinny (peaked)**.
- The data is clustered tightly around the mean.



Overall Revision Summary

- **Measure of Dispersion:** Checks how "spread out" your data is from the center (Mean).
- **Sample Formula Rule:** When calculating Sample Variance (s^2), always divide by $(n - 1)$, which is known as Bessel's Correction.
- **Standard Deviation (s):** It is simply the **square root** of the Variance ($\sqrt{s^2}$). It acts as a "step size" to find where data points are located left or right of the mean.
- **Graph Shape:**
 - **High SD:** Fat, wide, and flat curve. Data is far from the mean.
 - **Low SD:** Tall, skinny, and peaked curve. Data is super close to the mean.

Keep practicing! Understanding spread is just as important as understanding the center in Data Science!